

5D Parallelism for Large Model Training

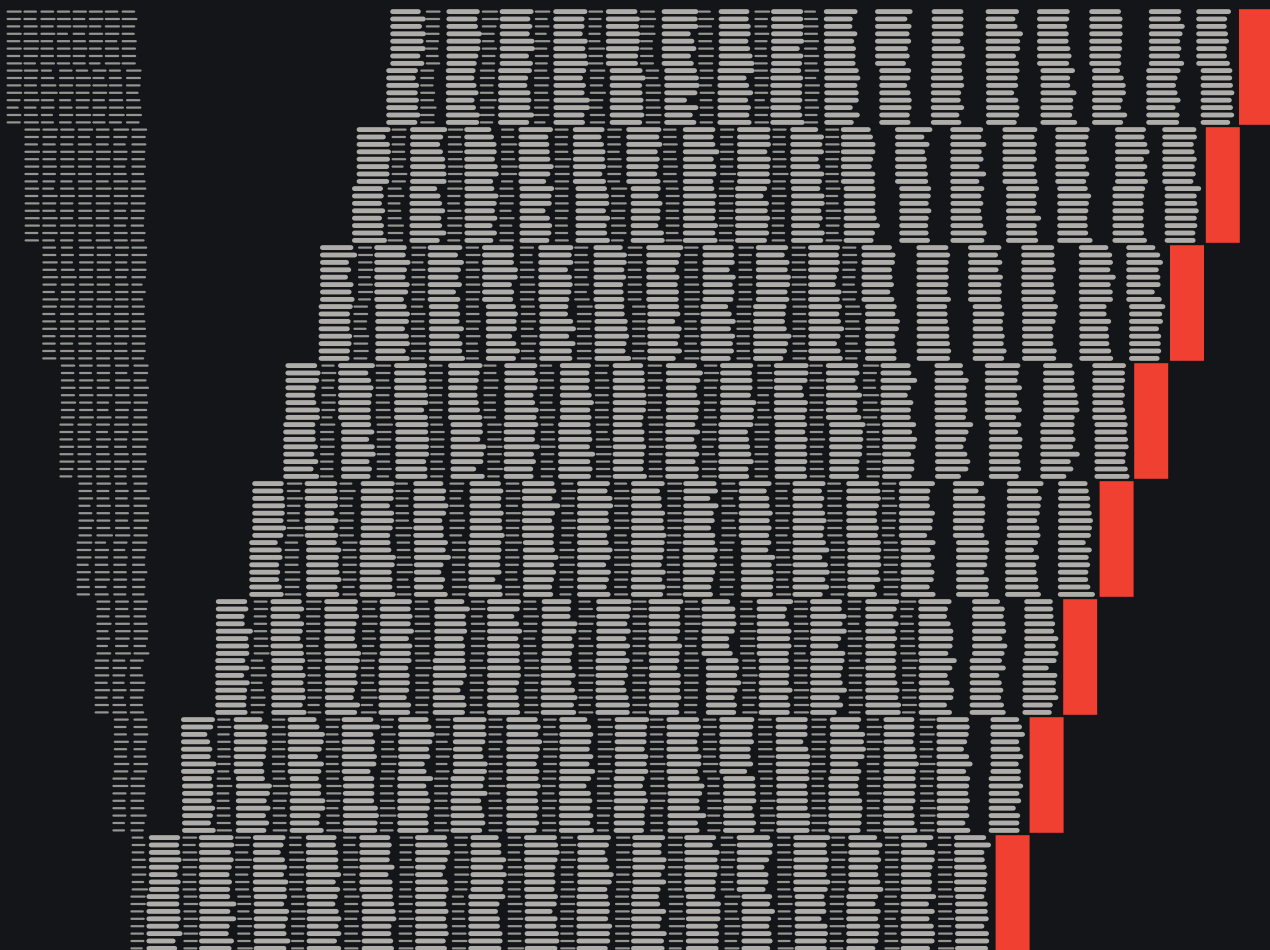


FIG. 0 One training step on a 128-GPU mesh — data $2 \times$ pipeline $8 \times$ tensor $2 \times$ context $2 \times$ expert 2 — simulated under a 1F1B schedule of 16 micro-batches with $\pm 7\%$ kernel-time noise. Each row is a GPU: thin = forward, thick = backward, red = gradient all-reduce.